

Maximize
 w, b

$$\frac{2}{\|w\|}$$

$\Rightarrow$

$$\boxed{\text{Minimize } w, b \quad \frac{\|w\|}{2}}$$

$$c_i = 1$$

cost function of SVM (SVC)

$$\text{Min}_{w, b} \quad \frac{\|w\|}{2}$$

$$+ \left[c_i \sum_{i=1}^n \xi_i \right] \rightarrow \boxed{\text{Hinge Loss}}$$

Hyperparameters

↓
Soft margin.

{ How many points we want to avoid misclassification }

{ Summation of the distance of the incorrect data point from the marginal plane }

